

Algorithms for Data Science

Lecture 2: Algorithms for Data Preprocessing





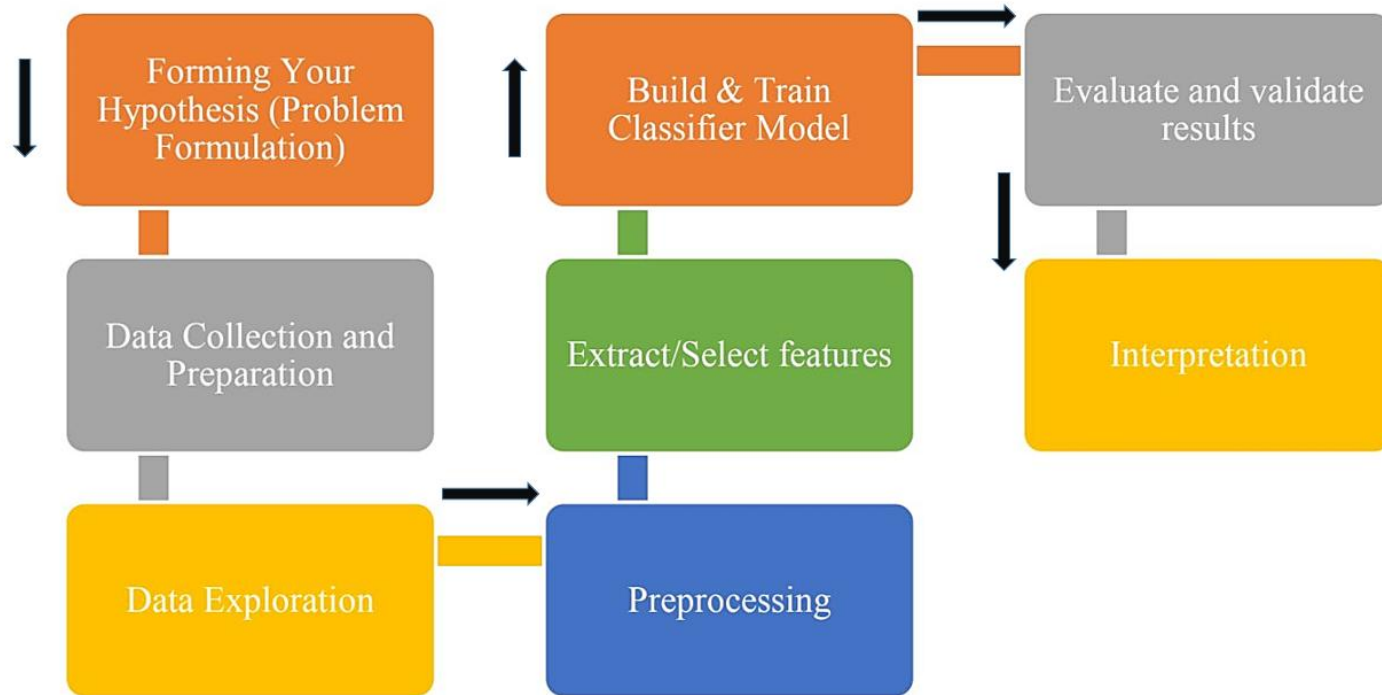
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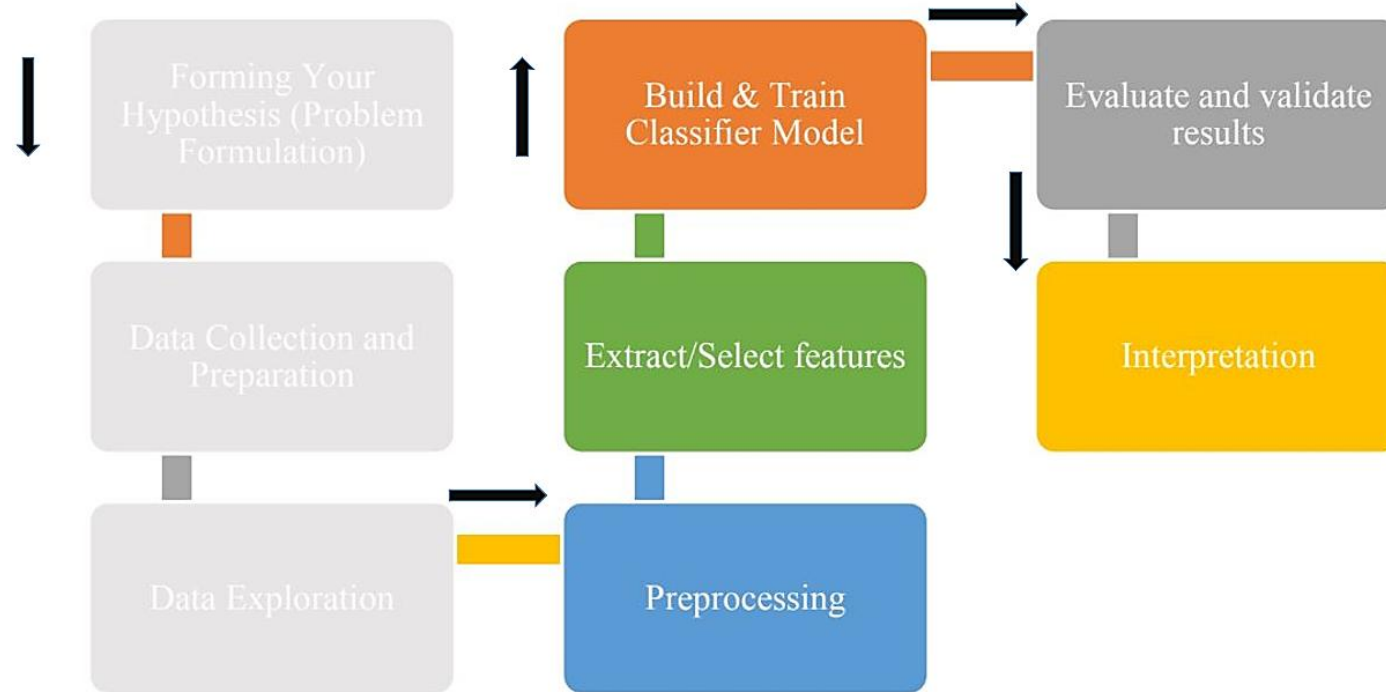


The Data Science Pipeline

In this course, we follow a structured pipeline:



The Data Science Pipeline



Key Objectives of the Preprocessing Phase

1. Data Cleaning

- ✓ Missing values handling
- ✓ Data Cleaning From Outlier

2. Data Transformation

3. Encoding Categorical Data



Preprocessing Phase- Data Cleaning

- Missing values handling
- Data Cleaning From Outlier



Preprocessing Phase- Data Cleaning

- **Handle Missing Values:** Missing data can also lead to misleading results by introducing bias.

Case	Age	Gender	Home	Education	Occupation
1	.	Female	No	16	Non-professional
2	22	Male	No	.	Non-professional
3	39	Male	.	20	Professional
4	.	Female	Yes	.	Professional
5	40	.	Yes	16	Non-professional
6	22	Female	No	16	.
7	35	Male	Yes	18	Professional
8	39	Male	Yes	20	Professional

- How to handle missing data ?



Preprocessing Phase- Data Cleaning

- **How to handle missing data ?**

1. Drop the row/column with missing data which is the easiest approach. However, some critical data points which may be important will be missing.
2. Filling missing values using mean or median value if it's a numerical variable.
3. Filling the missing data with mode, if it's a categorical value.
4. Imputation by K-NN: The object is assigned to the class with the most members among its k closest neighbors in the training set.



Descriptive Statistics

- **Descriptive statistics** are a set of brief descriptive coefficients that summarize a given data set, which can be either a representation of the entire population **or a sample of it**. Descriptive statistics includes the measures of **central tendency**.
- Measures of Central Tendency:
 - ✓ Mean
 - ✓ Mode
 - ✓ Median



Mean

- **Example (calculation of mean for binary data)**

The average for such data indicates the share of units in the sample. For example, for the following data:

1, 0, 0, 0, 1, 1, 1, 0, 0, 0

Mean = $4/10 = 0.4$

That is, 40% of the sample values take a value equal to one.



Mode

- The mode is the value of a variable that occurs more often than others. It is denoted by Mo . The mode can be defined on the data of any scale. There can be several values for the mode. If none of the values of the variable in the data set is repeated, it is said that there is no mode.

Example

Grade	Frequency
poor	9
sufficient	7
mode fair	14
good	9
excellent	1
Total	40



Median (1/3)

- It is a measure represents the middle value in a dataset when the values are arranged in ascending or descending order.
- **How to Calculate the Median:**
 1. **Arrange the Data:** Sort the dataset in ascending or descending order.
 2. **Find the Middle Value:** If the dataset has an odd number of observations, the median is the middle value. If the dataset has an even number of observations, the median is the average of the two middle values.



Median (2/3)

Formula:

- For a dataset with n values:
 - If n is odd:

$$\text{Median} = \left(\frac{n+1}{2} \right)^{\text{th}} \text{ value}$$

- If n is even:

$$\text{Median} = \frac{\left(\frac{n}{2} \right)^{\text{th}} \text{ value} + \left(\frac{n}{2} + 1 \right)^{\text{th}} \text{ value}}{2}$$



Median (3/3)

Example 1: Odd Number of Observations

Dataset: [3, 5, 1, 4, 2]

1. Sort the data: [1, 2, 3, 4, 5]
2. Since $n = 5$ (odd), the median is the 3rd value.

$$\text{Median} = 3$$

Example 2: Even Number of Observations

Dataset: [3, 5, 1, 4, 2, 6]

1. Sort the data: [1, 2, 3, 4, 5, 6]
2. Since $n = 6$ (even), the median is the average of the 3rd and 4th values.

$$\text{Median} = \frac{3 + 4}{2} = 3.5$$



What is the best measure of central tendency?

- We have considered 3 measures of the central tendency (Mean, Mode, Median). At first glance, it might seem that the mean is the most widely used measure. In terms of popularity, this is of course true, but in terms of applicability and usefulness, it is not always the best one, read the following example.
- A village has 50 inhabitants. Among them, 49 people are rural residents with a monthly income of 1 thousand rubles, and one resident is a prosperous farmer with an income of 451 thousand rubles. We calculate the average income of the villagers. It is equal to 10 thousand rubles. It is clear that this number does not adequately reflect the income of the villagers. In this case, it would be much more rational to use mode or median as a measure of the central tendency (both equal 1 thousand rubles).



Example 1

- Find the following descriptive statistics for the sample: (5,4,6,2,1,1,5,2,5,9,7)
 - ✓ Mode
 - ✓ Median
 - ✓ Mean



Example 1- Solution

- Find the following descriptive statistics for the sample: (5,4,6,2,1,1,5,2,5,9,7)

First of all, let's sort sample elements in ascending order (1,1,2,2,4,5,5,5,6,7,9)

- ✓ Mode = 5
- ✓ Median = 5
- ✓ Mean = 4.27



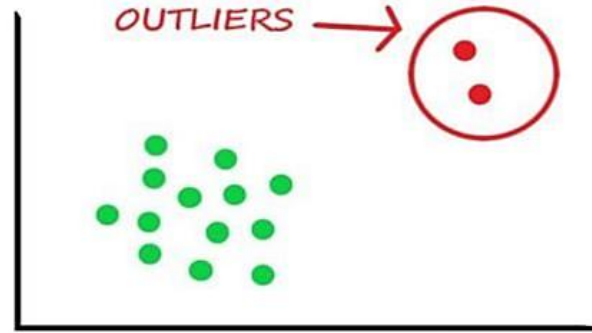
Preprocessing Phase- Data Cleaning

- Missing values handling
- Data Cleaning From Outlier



Data Cleaning From Outlier

Noise/Outlier data: are data points that deviate dramatically from the rest of the dataset's observations.



How outliers affects data ?

- The range and distribution of attribute values are important to machine learning algorithms.
- Outliers in the data can spoil and mislead the training process, resulting in longer training durations, less accurate models, and, ultimately, poorer results.



Data Cleaning From Outlier

- Outlier Detection Methods
 - ✓ Z-Score
 - ✓ Inter Quartile Range



Data Cleaning From Outlier - Important Terms

- **Variance:** Variance measures how far each number in a dataset is from the mean (average) and thus quantifies the spread of the data.

Formula:

- **Population Variance (σ^2):**

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$$

Where:

- N : Total number of data points.
- x_i : Each individual data point.
- μ : Population mean.



Data Cleaning From Outlier - Important Terms

- **Standard deviation** is the square root of the variance and provides a measure of dispersion in the same units as the original data.

Formula:

- **Population Standard Deviation (σ):**

$$\sigma = \sqrt{\sigma^2} = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2}$$



Data Cleaning From Outlier – (Z-Score)

- Z-Score is a statistical measure used to identify outliers in a dataset.
- To use the z-score for outlier detection, you first calculate the **mean and standard deviation of the dataset**.
- Then, for each observation in the dataset, you calculate the difference between the observation and the mean and divide it by the standard deviation. This gives you the z-score for that observation.



Data Cleaning From Outlier (Z-Score)

- The z-score tells you how many standard deviations away from the mean an observation is.
- Typically, a z-score greater than 3 or less than -3 is considered an outlier. This means that the observation is more than three standard deviations away from the mean and is therefore an unusual data point.
- By identifying outliers using the z-score, you can better understand the distribution of your data and make informed decisions about how to handle these unusual observations in your analysis.



Data Cleaning From Outlier - (Z-Score)

- **Outlier Detection Methods (Z-Score)**

Formula

The Z-score for a data point x is calculated as:

$$Z = \frac{(x - \mu)}{\sigma}$$

Where:

- x : The data point.
- μ : The mean of the dataset.
- σ : The standard deviation of the dataset.



Data Cleaning From Outlier- (Z-Score)

- **Example (1):** Given the dataset: {6, 8, 10, 12, 14} check outliers

Step 1: Calculate the Mean

Given the dataset: {6, 8, 10, 12, 14}

$$\text{Mean} = \frac{6 + 8 + 10 + 12 + 14}{5} = \frac{50}{5} = 10$$

Step 2: Calculate the Standard Deviation

$$\begin{aligned}\text{Standard Deviation} &= \sqrt{\frac{(6 - 10)^2 + (8 - 10)^2 + (10 - 10)^2 + (12 - 10)^2 + (14 - 10)^2}{5}} \\ &= \sqrt{\frac{(-4)^2 + (-2)^2 + 0^2 + 2^2 + 4^2}{5}} = \sqrt{\frac{16 + 4 + 0 + 4 + 16}{5}} = \sqrt{\frac{40}{5}} = \sqrt{8} \approx 2.83\end{aligned}$$



Data Cleaning From Outlier- (Z-Score)

- **Example (1):** Given the dataset: {6, 8, 10, 12, 14} check outliers

Step 3: Calculate Z-Scores

The z-score for each data point is calculated as:

$$z = \frac{X - \mu}{\sigma}$$

- $z_1 = \frac{6-10}{2.83} \approx -1.41$
- $z_2 = \frac{8-10}{2.83} \approx -0.71$
- $z_3 = \frac{10-10}{2.83} = 0$
- $z_4 = \frac{12-10}{2.83} \approx 0.71$
- $z_5 = \frac{14-10}{2.83} \approx 1.41$

With threshold $|z| > 3$, **no values are considered outliers** in this dataset.



Data Cleaning From Outlier- (Z-Score)

- Example (2): Given the dataset: {-20, 1, 1, 2, 2, 4, 5, 5, 6, 7, 18, 35} check outliers

Step 1: Calculate the Mean

$$\text{Mean} = \frac{-20 + 1 + 1 + 2 + 2 + 4 + 5 + 5 + 6 + 7 + 18 + 35}{12} = \frac{66}{12} = 5.5$$

Step 2: Calculate the Standard Deviation

$$\text{Standard Deviation} \approx 12.296$$



Data Cleaning From Outlier- (Z-Score)

- Example (2): Given the dataset: $\{-20, 1, 1, 2, 2, 4, 5, 5, 6, 7, 18, 35\}$ check outliers

Step 3: Calculate Z-Scores

$$\bullet z(-20) = \frac{-20-5.5}{12.296} \approx -2.11$$

$$\bullet z(1) = \frac{1-5.5}{12.296} \approx -0.372$$

$$\bullet z(1) = \frac{1-5.5}{12.296} \approx -0.372$$

$$\bullet z(2) = \frac{2-5.5}{12.296} \approx -0.29$$

$$\bullet z(2) = \frac{2-5.5}{12.296} \approx -0.29$$

$$\bullet z(4) = \frac{4-5.5}{12.296} \approx -0.124$$

$$\bullet z(5) = \frac{5-5.5}{12.296} \approx -0.041$$

$$\bullet z(6) = \frac{6-5.5}{12.296} \approx 0.041$$

$$\bullet z(7) = \frac{7-5.5}{12.296} \approx 0.124$$

$$\bullet z(18) = \frac{18-5.5}{12.296} \approx 1.033$$

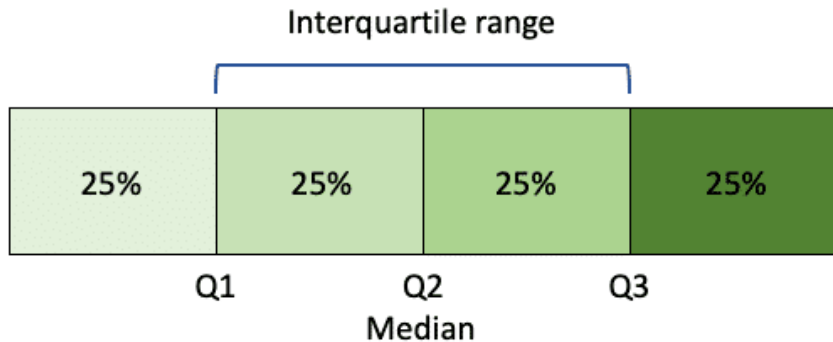
$$\bullet z(35) = \frac{35-5.5}{12.296} \approx 2.44$$



Data Cleaning From Outlier

- Outlier Detection Methods (Inter Quartile Range)

The Interquartile Range (IQR) is a measure of statistical dispersion that describes the spread of the middle 50% of a dataset. It is calculated as the difference between the third quartile (Q3) and the first quartile (Q1). The IQR is commonly used in the IQR test to identify outliers in a dataset.



Data Cleaning From Outlier– (IQR)

- After arrange the data in ascending order, follow the following steps

Steps to Calculate IQR and Detect Outliers

1. Calculate Quartiles

- **First Quartile (Q1):** The median of the lower half of the data (25th percentile).
- **Second Quartile (Q2):** The median of the dataset (50th percentile).
- **Third Quartile (Q3):** The median of the upper half of the data (75th percentile).

2. Calculate the Interquartile Range (IQR)

$$\text{IQR} = Q3 - Q1$$



Data Cleaning From Outlier– (IQR)

3. Define the Bounds:

- **Lower Bound:** $Q1 - k \cdot IQR$
- **Upper Bound:** $Q3 + k \cdot IQR$

4. Identify Outliers:

- Any data point **below the lower bound** or **above the upper bound** is considered a potential outlier.
-
- k is a constant, typically **1.5** for identifying mild outliers or **3** for identifying extreme outliers.



Data Cleaning From Outlier– (IQR)

- **Moderate Outlier** ones are those outliers that are located below the first quartile or above the third quartile for the distance of more than **1.5 IQR**, but less than 3 IQR.
- **Extreme outliers** are located below the first quartile or above the third one by the distance of more than **3 IQR**.



Data Cleaning From Outlier– (IQR)

- **Example (1): Given the following data find the moderate outliers.**

Dataset:

$\{4, 5, 1, 2, 3, 8, 9, 20, 6, 7\}$

Arrange the Data in Ascending Order

$\{1, 2, 3, 4, 5, 6, 7, 8, 9, 20\}$



Data Cleaning From Outlier– (IQR)

- **Example (1)**

Step 1: Calculate Q1 and Q3

- **Q1 (First Quartile):** Median of the lower half of the data.

Lower half = $\{1, 2, 3, 4, 5\}$

$$Q1 = 3$$

- **Q3 (Third Quartile):** Median of the upper half of the data.

Upper half = $\{6, 7, 8, 9, 20\}$

$$Q3 = 8$$



Data Cleaning From Outlier– (IQR)

- **Example (1)**

Step 2: Calculate the Interquartile Range (IQR)

$$\text{IQR} = Q3 - Q1 = 8 - 3 = 5$$

Step 3: Determine the Outlier Boundaries

- **Lower Bound:**

$$\text{Lower Bound} = Q1 - 1.5 \times \text{IQR} = 3 - 1.5 \times 5 = -4.5$$

- **Upper Bound:**

$$\text{Upper Bound} = Q3 + 1.5 \times \text{IQR} = 8 + 1.5 \times 5 = 15.5$$



Data Cleaning From Outlier– (IQR)

- **Example (2):** find Moderate and Extreme outliers in the following dataset:

-20,1,1,2,2,4,5,5,6,7,18,35.

Step 1: Calculate Q1, Q3, and IQR

- **Q1 (First Quartile):** 1.5
- **Q3 (Third Quartile):** 6.5
- **IQR (Interquartile Range):**

$$\text{IQR} = \text{Q3} - \text{Q1} = 6.5 - 1.5 = 5$$



Data Cleaning From Outlier– (IQR)

- **Example (2)**

Step 2: Determine Moderate Outliers

- **Lower Bound for Moderate Outliers:**

$$Q1 - 1.5 \times IQR = 1.5 - 1.5 \times 5 = -6$$

- **Upper Bound for Moderate Outliers:**

$$Q3 + 1.5 \times IQR = 6.5 + 1.5 \times 5 = 14$$

- **Moderate Outliers:**

- Any value below **-6** or above **14**.
- In this dataset:

$$\text{Moderate Outliers} = \{-20, 18, 35\}$$



Data Cleaning From Outlier– (IQR)

- **Example (2)**

Step 3: Determine Extreme Outliers

- **Lower Bound for Extreme Outliers:**

$$Q1 - 3 \times IQR = 1.5 - 3 \times 5 = -13.5$$

- **Upper Bound for Extreme Outliers:**

$$Q3 + 3 \times IQR = 6.5 + 3 \times 5 = 21.5$$

- **Extreme Outliers:**

- Any value below **-13.5** or above **21.5**.
- In this dataset:

$$\text{Extreme Outliers} = \{-20, 35\}$$



Data Cleaning From Outlier– (IQR)

- **Example (3):** Find Moderate and Extreme outliers in the following dataset:

Dataset: {0,1,2,4,5,5,7,10,10,12,13,17,39}

Step 1: Calculate Q1, Q3, and IQR

- **Q1 (First Quartile):** 3
- **Q3 (Third Quartile):** 12.5
- **IQR (Interquartile Range):**

$$\text{IQR} = Q3 - Q1 = 12.5 - 3 = 9.5$$



Data Cleaning From Outlier– (IQR)

- **Example (3)**

Step 2: Determine Moderate Outliers

- **Lower Bound for Moderate Outliers:**

$$Q1 - 1.5 \times IQR = 3 - 1.5 \times 9.5 = -11.25$$

- **Upper Bound for Moderate Outliers:**

$$Q3 + 1.5 \times IQR = 12.5 + 1.5 \times 9.5 = 26.75$$

- **Moderate Outliers:**

- Any value below **-11.25** or above **26.75**.
- In this dataset:

$$\text{Moderate Outliers} = \{39\}$$



Data Cleaning From Outlier– (IQR)

- **Example (3)**

Step 3: Determine Extreme Outliers

- **Lower Bound for Extreme Outliers:**

$$Q1 - 3 \times IQR = 3 - 3 \times 9.5 = -25.5$$

- **Upper Bound for Extreme Outliers:**

$$Q3 + 3 \times IQR = 12.5 + 3 \times 9.5 = 41$$

- **Extreme Outliers:**

- Any value below **-25.5** or above **41**.
- In this dataset:

Extreme Outliers = {} (No extreme outliers)



Key Objectives of the Preprocessing Phase

1. Data Cleaning
2. Data Transformation
3. Encoding Categorical Data



Preprocessing Phase- Data Transformation

- **Data transformation** refers to the process of modifying the format, structure, or values of data to make it more usable and meaningful.
- The Russian national exam result (RNE) example shows different ranges of values.

Subject	Number of Top-Rated Students
Russian Language	3722
Mathematics (Profile)	145
History	206
Chemistry	634
English Language	15
Informatics and ICT	254
Geography	64
French Language	2



Preprocessing Phase- Data Transformation

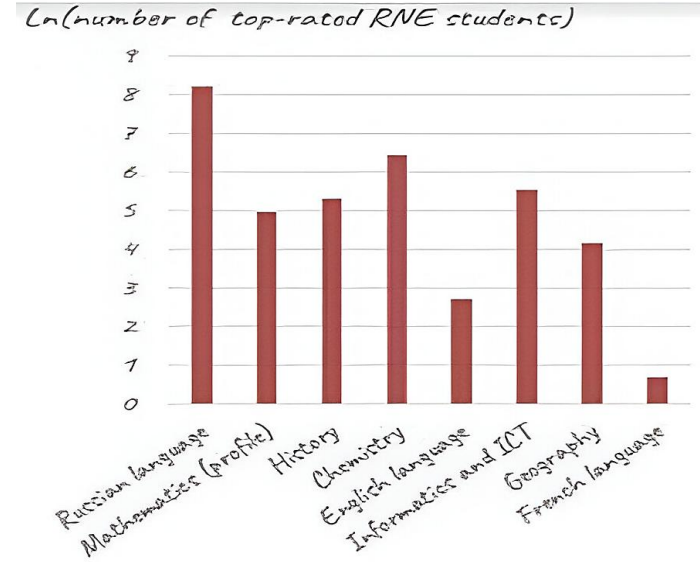
- Natural or decimal logarithm is suitable for data conversion, keeps order among values, but is not appropriate if there are zero values in the source data.
- The square root transformation also maintains the order between values and is appropriate for zero values, but not for negative values.



Preprocessing Phase- Data Transformation

- Let's consider how the transformations are applied on the example of a logarithmic transformation. Logarithmic transformations have a significant effect on the form of distribution.

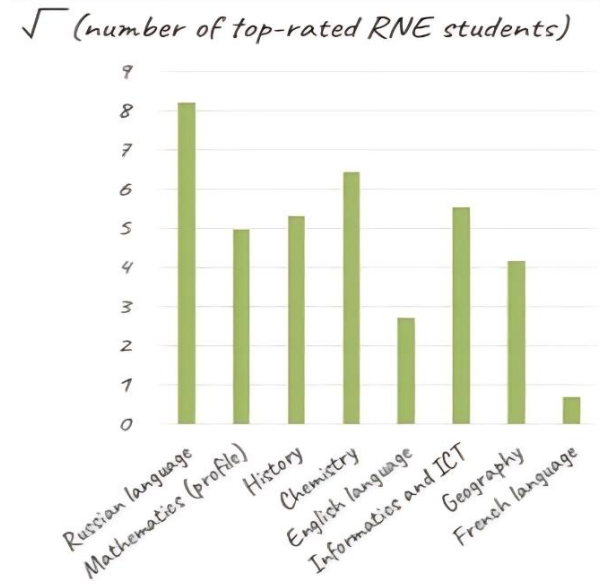
Subject	Number of Top-Rated RNE Students	$\ln(\text{Number of Top-Rated RNE Students})$
Russian Language	3722	8.22
Mathematics (Profile)	145	4.98
History	206	5.33
Chemistry	634	6.45
English Language	15	2.77
Informatics and ICT	254	5.54
Geography	64	4.16
French Language	2	0.69



Preprocessing Phase- Data Transformation

- The transformation of the square root has a more moderate effect on the shape of the distribution. The following graph shows the bar chart of top-rated RNE students after applying the square root transformation.

Subject	Number of Top-Rated RNE Students	$\sqrt{\text{Number of Top-Rated RNE Students}}$
Russian Language	3722	61.01
Mathematics (Profile)	145	12.04
History	206	14.35
Chemistry	634	25.18
English Language	15	3.87
Informatics and ICT	254	15.94
Geography	64	8.00
French Language	2	1.41



Preprocessing Phase- Data Transformation

- **How to choose a transformation type?**

There are many possible transformations. How to choose the appropriate one? The answer to this question is not obvious, although there are formal statistical methods to choose the transformation. If you do not delve into these theories, then the general strategy for choosing a transformation is to clearly define the purpose of the transformation (for example, visualization of a certain type), apply the most commonly used transformations such as logarithms, square root then choose the best method based on the purpose and results.

Transformation Limitations

Transformation	Cannot Be Applied To
$\ln(x)$	- Zero values
$\sqrt{x}$ (Square Root)	- Negative values
$\log_{10}(x)$	- Zero values
$1/x$	- Zero values



Preprocessing Phase- Data Transformation

- **Data normalization** is a preprocessing technique used to scale data to **a specific range or distribution**. It is essential for ensuring that features contribute equally to machine learning models and improving the performance of algorithms that are sensitive to the scale of data, such as gradient descent-based models, k-Nearest Neighbors (k-NN), and Principal Component Analysis (PCA).
- **The key methods of data normalization:**
 - ✓ Min-Max Normalization
 - ✓ Z-Score Standardization
 - ✓ Robust Scaling



Preprocessing Phase- Data Transformation

- Data normalization
 - ✓ **Min-Max Normalization:** Scales data to a fixed range, typically [0, 1].

Formula:

$$X_{\text{normalized}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

Example:

- Original data: [10, 20, 30, 40, 50]
- $X_{\min} = 10, X_{\max} = 50$
- Normalized data:

$$\frac{10 - 10}{50 - 10} = 0, \quad \frac{20 - 10}{50 - 10} = 0.25, \quad \frac{30 - 10}{50 - 10} = 0.5, \quad \frac{40 - 10}{50 - 10} = 0.75, \quad \frac{50 - 10}{50 - 10} = 1$$

Result: [0, 0.25, 0.5, 0.75, 1]



Preprocessing Phase- Data Transformation

- **Data normalization**

- ✓ **Z-Score Standardization:** Scales data to have a mean of 0 and a standard deviation of 1.

Formula:

$$X_{\text{standardized}} = \frac{X - \mu}{\sigma}$$

- μ : Mean of the data.
- σ : Standard deviation of the data.

Example:

- **Original data:** [10, 20, 30, 40, 50]
- **Mean (μ):** 30
- **Standard deviation (σ):** 15.81



Preprocessing Phase- Data Transformation

- Data normalization

- ✓ **Z-Score Standardization:** Scales data to have a mean of 0 and a standard deviation of 1.

Standardized Data:

$$\frac{10 - 30}{15.81} = -1.26, \quad \frac{20 - 30}{15.81} = -0.63, \quad \frac{30 - 30}{15.81} = 0,$$
$$\frac{40 - 30}{15.81} = 0.63, \quad \frac{50 - 30}{15.81} = 1.26$$

Result:

$$[-1.26, -0.63, 0, 0.63, 1.26]$$



Preprocessing Phase- Data Transformation

- **Data normalization**

- ✓ **Robust Scaling:** Scales data using the interquartile range (IQR), making it robust to outliers.

Formula:

$$X_{\text{scaled}} = \frac{X - \text{Median}}{\text{IQR}}$$

Where:

- x : The original data value.
- **Median:** The median of the dataset.
- **IQR:** The interquartile range ($Q3 - Q1$).



Preprocessing Phase- Data Transformation

- **Data normalization**
 - ✓ **Robust Scaling:** Scales data using the interquartile range (IQR), making it robust to outliers.

Steps to Calculate Robust Scaling

1. **Arrange the data in ascending order.**
2. **Calculate the median** of the dataset.
3. **Calculate the first quartile (Q1) and third quartile (Q3).**
4. **Compute the IQR:** $IQR = Q3 - Q1$.
5. **Apply the robust scaling formula** to each data point.



Preprocessing Phase- Data Transformation

- Data normalization

- ✓ **Robust Scaling:** Scales data using the interquartile range (IQR), making it robust to outliers.

Example: Robust Scaling for the Dataset 10, 20, 30, 40, 50

Step 1: Arrange the data in ascending order

The data is already in ascending order:

10, 20, 30, 40, 50

Step 2: Calculate the median

The median is the middle value of the dataset. For this dataset:

Median = 30



Preprocessing Phase- Data Transformation

- **Data normalization**

- ✓ **Robust Scaling:** Scales data using the interquartile range (IQR), making it robust to outliers.

Step 3: Calculate Q1 and Q3

- **Q1:** Median of the lower half of the data (values below the median).
 - Lower half: 10, 20
 - $Q1 = \frac{10+20}{2} = 15$
- **Q3:** Median of the upper half of the data (values above the median).
 - Upper half: 40, 50
 - $Q3 = \frac{40+50}{2} = 45$

Step 4: Calculate the IQR

$$IQR = Q3 - Q1 = 45 - 15 = 30$$



Preprocessing Phase- Data Transformation

- Data normalization

- ✓ **Robust Scaling:** Scales data using the interquartile range (IQR), making it robust to outliers.

Step 5: Apply the robust scaling formula

Now, scale each value in the dataset using the formula:

$$\text{Scaled Value} = \frac{x - \text{Median}}{\text{IQR}}$$

1. For 10:

$$\frac{10 - 30}{30} = \frac{-20}{30} = -0.6667$$

2. For 20:

$$\frac{20 - 30}{30} = \frac{-10}{30} = -0.3333$$

3. For 30:

$$\frac{30 - 30}{30} = \frac{0}{30} = 0$$

4. For 40:

$$\frac{40 - 30}{30} = \frac{10}{30} = 0.3333$$

5. For 50:

Final Scaled Data:

-0.6667, -0.3333, 0, 0.3333, 0.6667

Preprocessing Phase- Data Transformation

- Data normalization

When to Use Which Method?

1. Min-Max Normalization:

- Use when data needs to be scaled to a specific range (e.g., $[0, 1]$).
- Sensitive to outliers.

2. Z-Score Standardization:

- Use when data needs to have a mean of 0 and a standard deviation of 1.
- Suitable for algorithms that assume normally distributed data.

3. Robust Scaling:

- Use when data contains outliers, as it is less sensitive to extreme values.



Key Objectives of the Preprocessing Phase

1. Data Cleaning
2. Data Transformation
3. Encoding Categorical Data



Preprocessing Phase- Encoding Categorical Data

- **Encoding categorical data** is the process of converting categorical variables (non-numeric data) into numerical format so that machine learning algorithms can process them. Most algorithms require numerical input, so encoding is a crucial step in data preprocessing.
- **The key methods for encoding categorical data:**
 - ✓ Label Encoding
 - ✓ One-Hot Encoding
 - ✓ Binary Encoding



Preprocessing Phase- Encoding Categorical Data

- **Label Encoding:** It Assigns a unique integer to each category.

Example:

- **Original Data:** ["Red", "Green", "Blue", "Green", "Red"]
- **Encoded Data:** [0, 1, 2, 1, 0]



Preprocessing Phase- Encoding Categorical Data

- **One-Hot Encoding:** Converts each category into a binary vector.

Example:

- **Original Data:** ["Red", "Green", "Blue", "Green", "Red"]
- **Encoded Data:**
 - Red: [1, 0, 0]
 - Green: [0, 1, 0]
 - Blue: [0, 0, 1]



Preprocessing Phase- Encoding Categorical Data

- **Binary Encoding:** Converts categories into binary code and splits the binary digits into separate columns.

Example:

- **Original Data:** ["A", "B", "C", "D"]
- **Encoded Data:**
 - A: 00 → [0, 0]
 - B: 01 → [0, 1]
 - C: 10 → [1, 0]
 - D: 11 → [1, 1]



Any Question?

